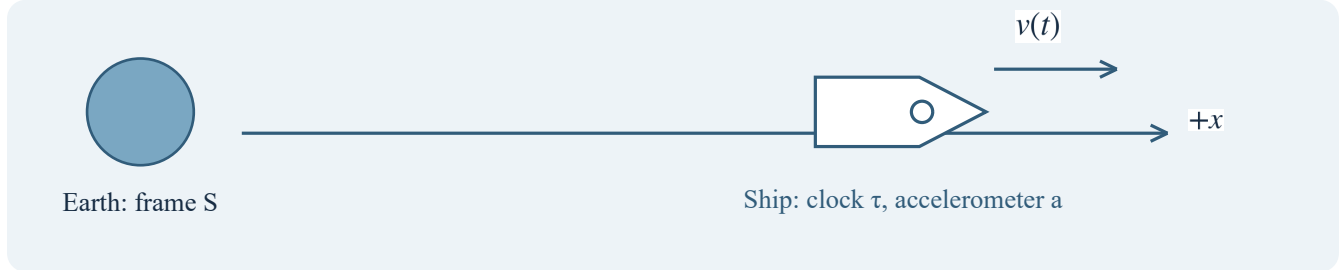


1. Scenario and the shared equation

Constant proper acceleration: establish the infinitesimal velocity change.

A spacecraft starts from rest beside Earth and travels along $+x$. Its onboard accelerometer reads a constant acceleration a . Let t denote Earth time, τ the ship clock reading, and v the ship velocity relative to Earth.



$$t = \tau = 0, \quad x = 0, \quad v = 0$$

Three frames during one infinitesimal step

Over the Earth-time interval dt , distinguish three inertial frames:

Earth frame S : the ship velocity changes from v to $v + dv$.

Initial instantaneous rest frame S_0 : comoving with the ship at the beginning; its velocity relative to Earth is v .

Final instantaneous rest frame S_1 : comoving with the ship at the end; its velocity relative to Earth is $v + dv$.

Keep S_0 fixed during the step. The ship starts at rest in S_0 and gains an infinitesimal velocity du in that frame. Thus du is also the velocity of S_1 relative to S_0 . The accelerometer measures this local change per ship-clock interval:

$$du = a d\tau$$

Compose the velocity of S_0 relative to Earth with that of S_1 relative to S_0 , using the velocity-addition formula:

$$v + dv = \frac{v + du}{1 + \frac{v du}{c^2}}$$

Expand the denominator to first order in du :

$$\begin{aligned} \left(1 + \frac{v du}{c^2}\right)^{-1} &= 1 - \frac{v du}{c^2} + O\left(\left(\frac{du}{c}\right)^2\right) \\ v + dv &\simeq (v + du)\left(1 - \frac{v du}{c^2}\right) \simeq v + du - \frac{v^2}{c^2} du \end{aligned}$$

Discard the terms quadratic in du , subtract v , and substitute $du = a d\tau$. In differential form,

$$dv = \left(1 - \frac{v^2}{c^2}\right) a d\tau \quad \implies \quad \frac{dv}{d\tau} = a \left(1 - \frac{v^2}{c^2}\right)$$

Choose the clock used to describe the motion:

2. Ship-clock branch: find $v(\tau)$.

3. Earth-clock branch: find $v(t)$.

2. Ship-clock branch

Integrate the shared equation with respect to the ship clock.

Use τ as the independent variable. Here v remains the ship velocity relative to Earth; the clock used to parametrize that velocity is the ship clock. Separate variables and apply $v=0$ at $\tau=0$:

$$\int_0^v \frac{dv'}{1 - \frac{v'^2}{c^2}} = a \int_0^\tau d\tau' = a\tau$$

Integrate by partial fractions

Factor the denominator and decompose the velocity integrand:

$$\frac{1}{1 - \frac{v'^2}{c^2}} = \frac{c^2}{(c - v')(c + v')} = \frac{c}{2} \left(\frac{1}{c - v'} + \frac{1}{c + v'} \right)$$

Integrating the two terms produces logarithms. For $0 \leq v < c$, both logarithm arguments are positive:

$$\int_0^v \frac{dv'}{1 - \frac{v'^2}{c^2}} = \frac{c}{2} \left[\ln \left(\frac{c + v'}{c - v'} \right) \right]_0^v$$

$$\frac{c}{2} \ln \left(\frac{c + v}{c - v} \right) = a\tau$$

Solve the logarithmic equation for velocity

Multiply by $2/c$ and exponentiate:

$$\frac{c + v}{c - v} = e^{\frac{2a\tau}{c}}$$

Multiply by $c - v$, then collect the terms containing v :

$$c + v = c e^{\frac{2a\tau}{c}} - v e^{\frac{2a\tau}{c}}$$

$$(1 + e^{\frac{2a\tau}{c}})v = c(e^{\frac{2a\tau}{c}} - 1)$$

Divide to isolate v . Multiplying the numerator and denominator by $e^{-a\tau/c}$ gives the defining exponential ratio of the hyperbolic tangent:

$$v = c \frac{e^{\frac{2a\tau}{c}} - 1}{e^{\frac{2a\tau}{c}} + 1} = c \frac{e^{\frac{a\tau}{c}} - e^{-\frac{a\tau}{c}}}{e^{\frac{a\tau}{c}} + e^{-\frac{a\tau}{c}}}$$

$$v(\tau) = c \tanh\left(\frac{a\tau}{c}\right)$$

3. Earth-clock branch

Convert the time variable, then use the cosine substitution.

The two nearby events lie on the ship's worldline, so the ship-clock interval is proper time. Time dilation relates it to Earth time:

$$dt = \gamma d\tau, \quad \gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

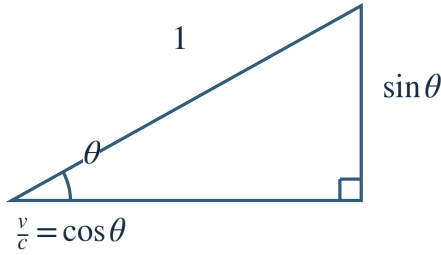
Apply the chain rule to the shared differential equation:

$$\frac{dv}{dt} = \frac{dv}{d\tau} \frac{d\tau}{dt} = a \left(1 - \frac{v^2}{c^2}\right)^{\frac{1}{\gamma}} = a \left(1 - \frac{v^2}{c^2}\right)^{3/2}$$

$$\int_0^v \frac{dv'}{\left(1 - \frac{v'^2}{c^2}\right)^{3/2}} = a \int_0^t dt' = at$$

Trigonometric substitution

Set $v/c = \cos \theta$. The triangle has hypotenuse 1, adjacent side v/c , and opposite side $\sin \theta = \sqrt{1 - v^2/c^2}$.



$$\frac{v}{c} = \cos \theta$$

$$dv = -c \sin \theta d\theta$$

Under this substitution, $dv = -c \sin \theta d\theta$ and the denominator becomes $\sin^3 \theta$. Thus an antiderivative is

$$\begin{aligned} \int \frac{dv}{\left(1 - \frac{v^2}{c^2}\right)^{3/2}} &= -c \int \frac{\sin \theta}{\sin^3 \theta} d\theta = -c \int \csc^2 \theta d\theta \\ &= c \cot \theta + C = c \frac{\cos \theta}{\sin \theta} + C = \frac{v}{\sqrt{1 - \frac{v^2}{c^2}}} + C \end{aligned}$$

Evaluate at the velocity limits. The lower endpoint $v=0$ contributes zero:

$$\left[\frac{v'}{\sqrt{1 - \frac{v'^2}{c^2}}} \right]_0^v = at \implies \frac{v}{\sqrt{1 - \frac{v^2}{c^2}}} = at$$

Square and collect the terms in v^2 . Choose the positive root for motion along $+x$:

$$\begin{aligned} v^2 &= a^2 t^2 \left(1 - \frac{v^2}{c^2}\right) \implies \left(1 + \frac{a^2 t^2}{c^2}\right) v^2 = a^2 t^2 \\ v(t) &= \frac{at}{\sqrt{1 + \left(\frac{at}{c}\right)^2}} \end{aligned}$$